

Firefly: An Interactive Browser-Based Vis Tool for Particle Data

Aaron M. Geller

Lead Data Scientist

Research Computing and Data Services

Northwestern

INFORMATION TECHNOLOGY
RESEARCH COMPUTING AND DATA SERVICES

The background of the slide is a deep space image featuring a bright yellow star in the center, surrounded by a glowing orange ring of dust or gas. The background is filled with numerous small, distant stars.

WHY we
(should)
visualize our
data

***Why*: Data Expl-oration / -anation**

Exploration

Help YOU learn about your data

Variety of formats, particularly important with multi-dimensional and complex data.

Explanation

Help OTHERS learn about your data

Typically, scientific figures in papers, posters, presentations, etc.

***Why*: Data Expl-oration / -anation**

Exploration

Help YOU learn about your data

Variety of formats, particularly important with multi-dimensional and complex data.

Explanation

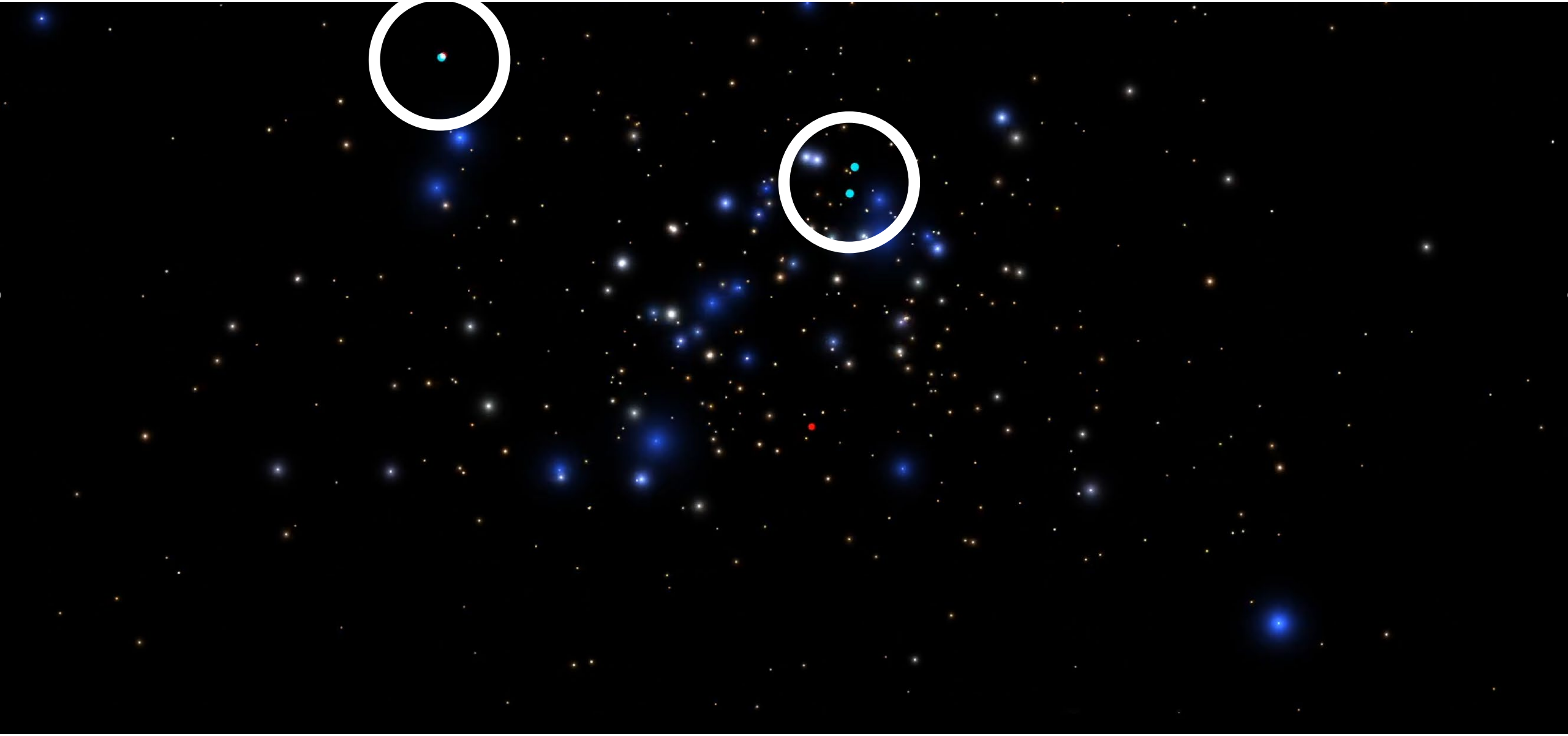
Help OTHERS learn about your data

Typically, scientific figures in papers, posters, presentations, etc.





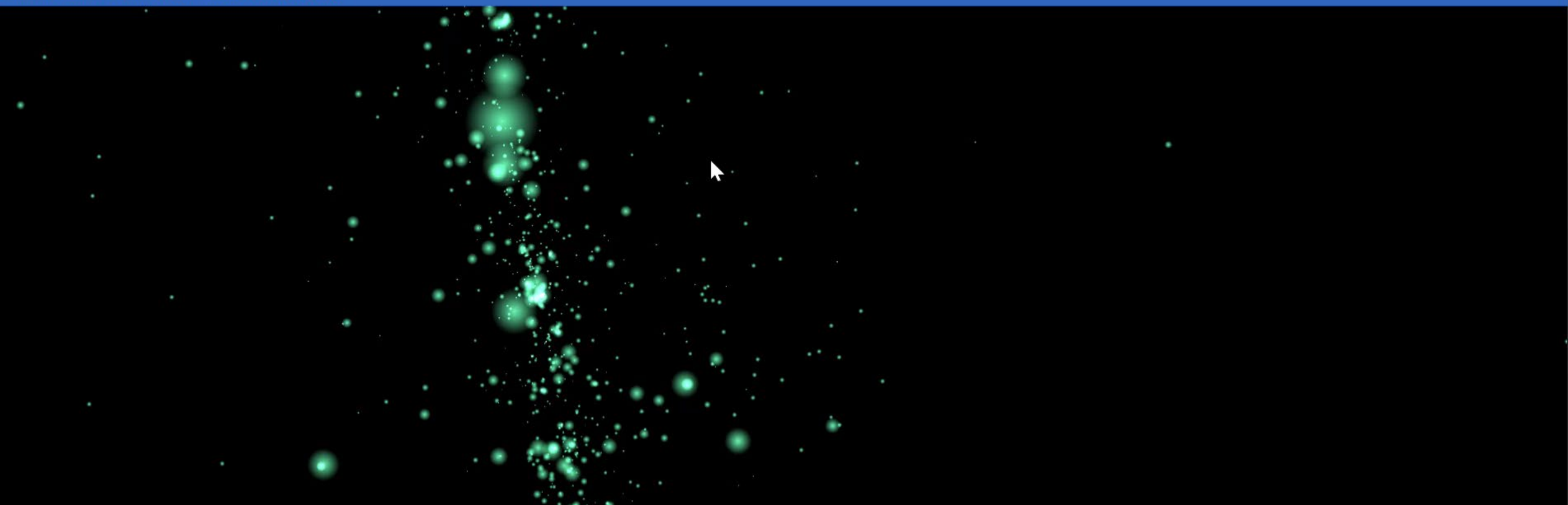
***Why* : Scientific discovery!**



Open Cluster Binary Explorer

Access photometric binary-star data in a collection of open clusters

An NSF-funded research project from PI [Aaron M. Geller](#).



We use the Bayesian Analysis of Stellar Evolution with Nine Parameters ([BASE-9](#)) software suite along with Gaia kinematics and distances and photometry from Gaia, Pan-STARRS and 2MASS to characterise the binary-star populations in a collection of open clusters. For information about our methods and results, please see the Papers section at the bottom of this page. You can access these data by clicking on the button below.

The interactive visualization above shows the open cluster population as seen from Earth, with the clusters in this study highlighted in pink.

Click here to enter the *Interactive Data Explorer*

View, filter, sort, create, edit, and download data and plots

1.0 Myr
1.1 pc
46.7 ms (21 fps)

Initial Conditions

N

Vel

Rho

D

Rg

ecc

Sizes

Nml

BH

NS

WD

Time

dt

Str

Res

Unity

GPGPU Star Cluster Viz

Aaron Geller

<https://github.com/ageller/UnityNbodyCluster>

0.0 Myr
5.0 pc
21.1 ms (47 fps)

Initial Conditions

N

Vel

R_{hm}

D

R_g

ecc

Sizes

N_{ml}

BH

NS

WD

Time

dt

Start

Restart

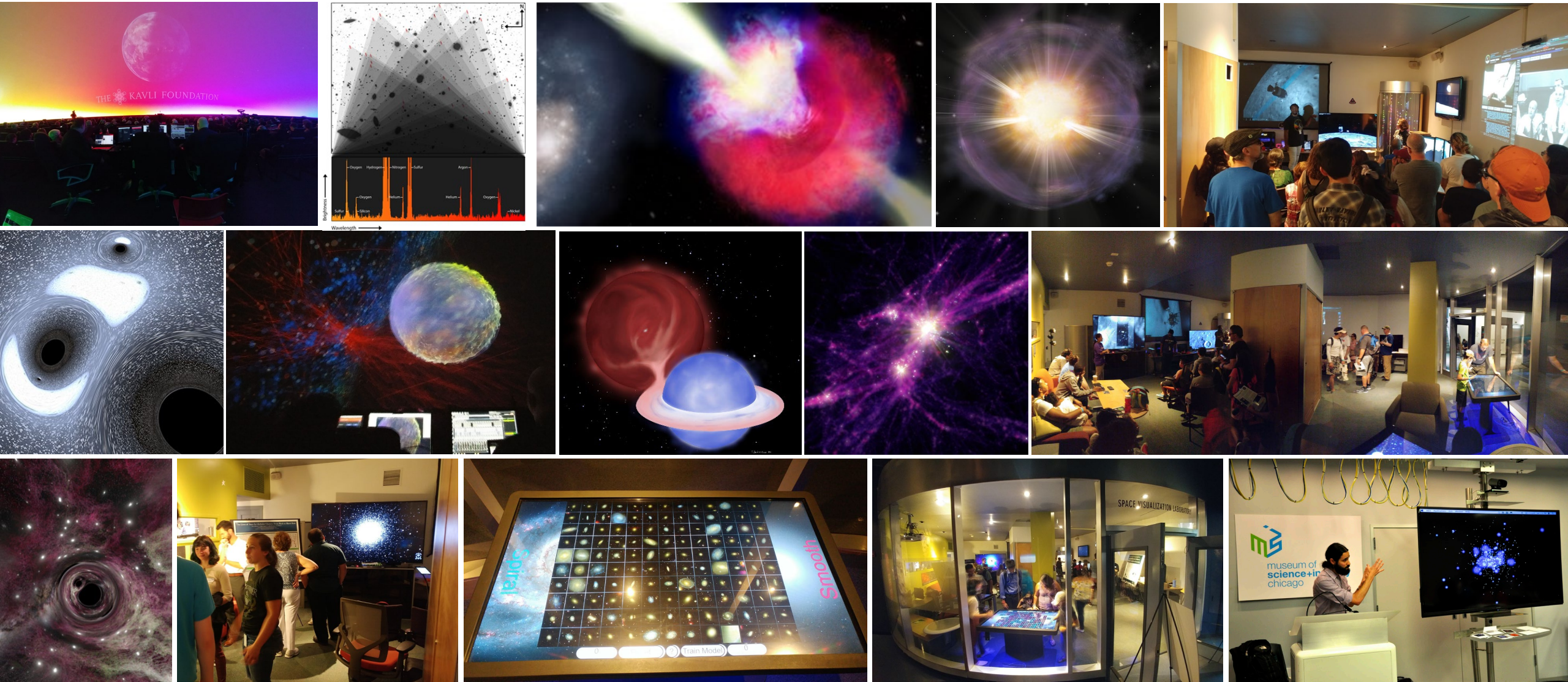
***Why*: Bug checking**

NGC 2682

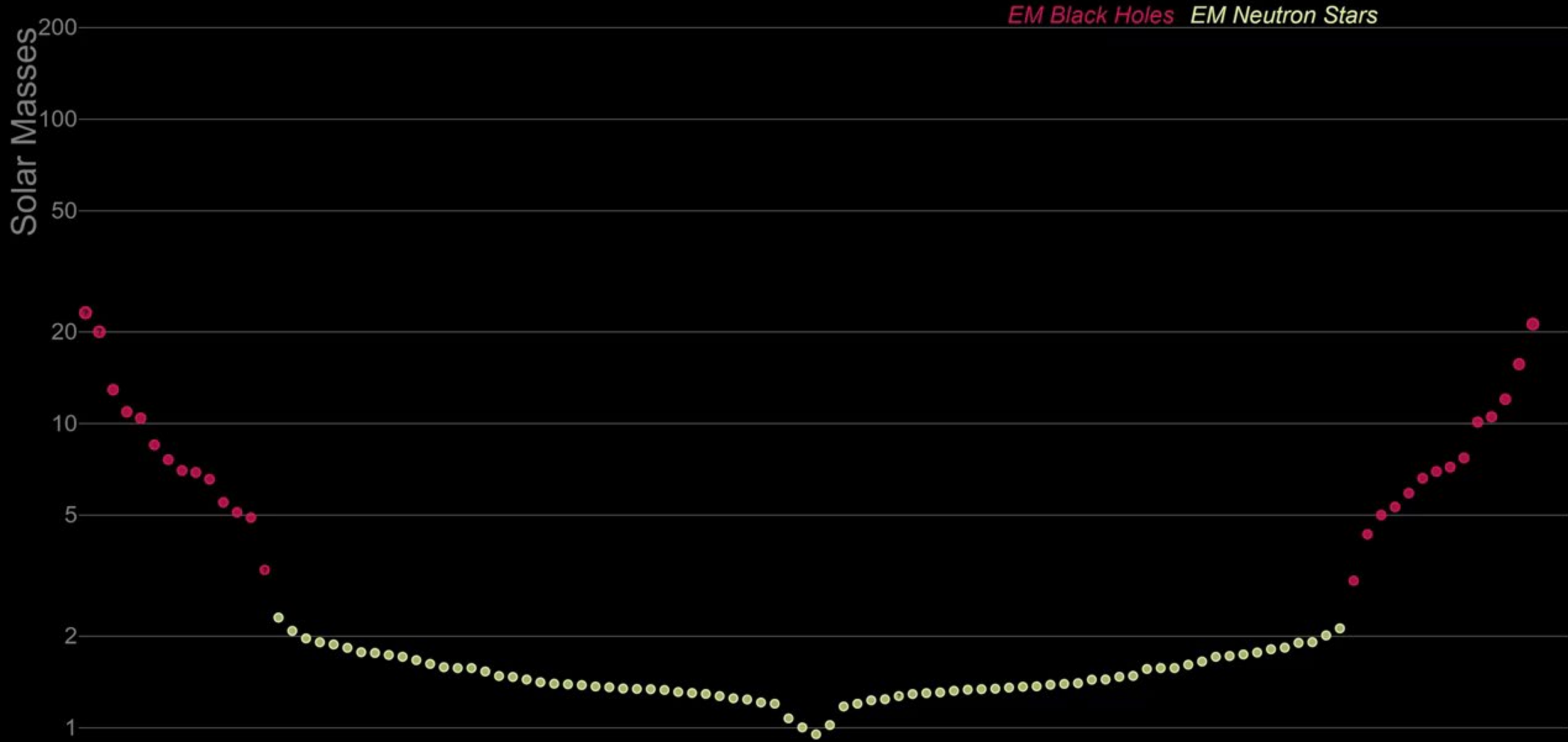


Why: Public Engagement

In person and online / press

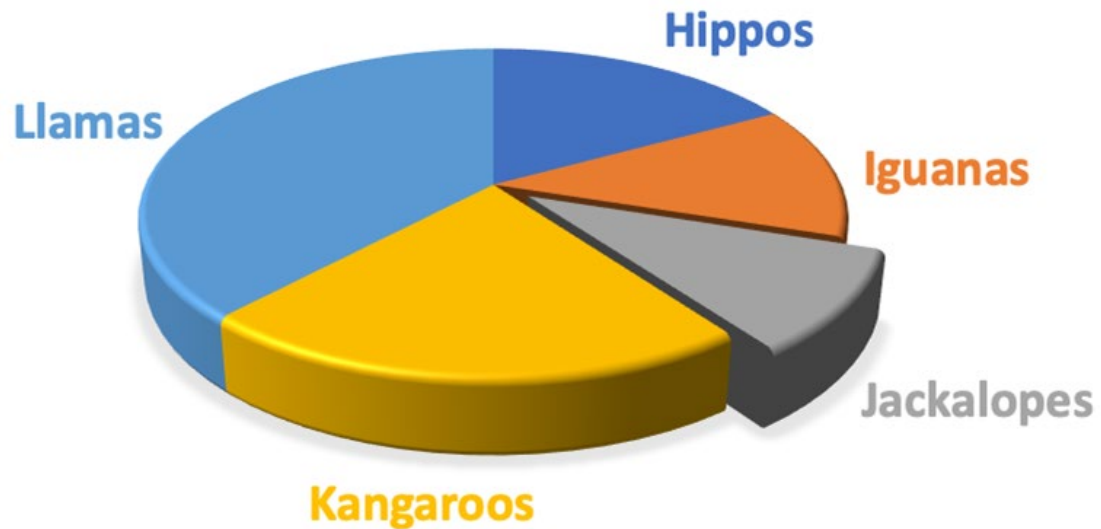


Masses in the Stellar Graveyard

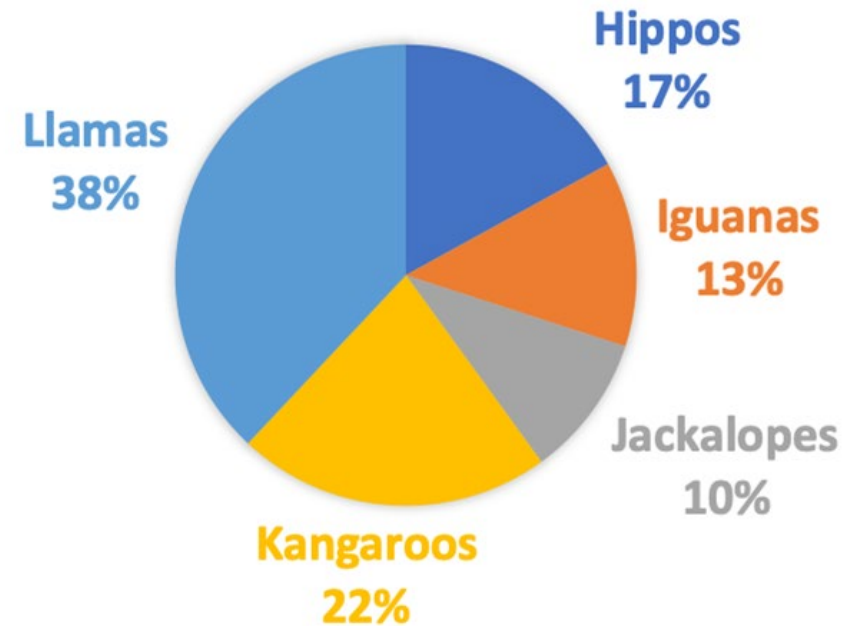


Visualizing Data in 3D ???

Stick to 2D whenever possible.



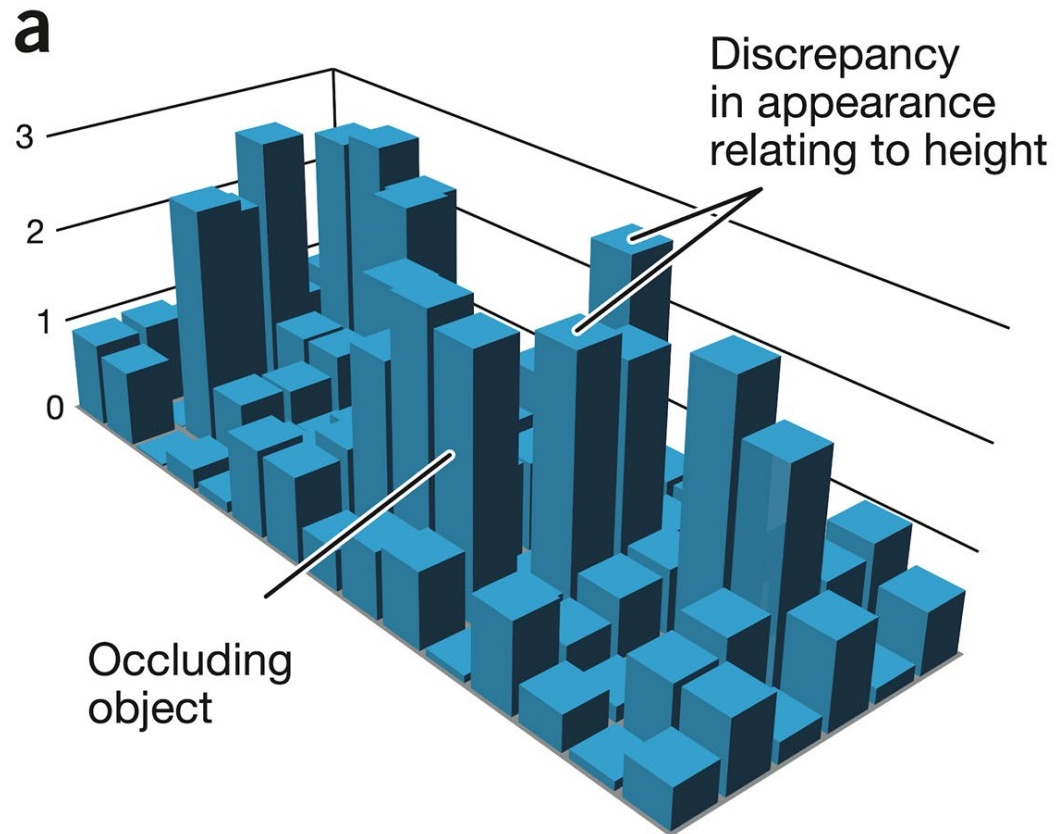
No



Yes

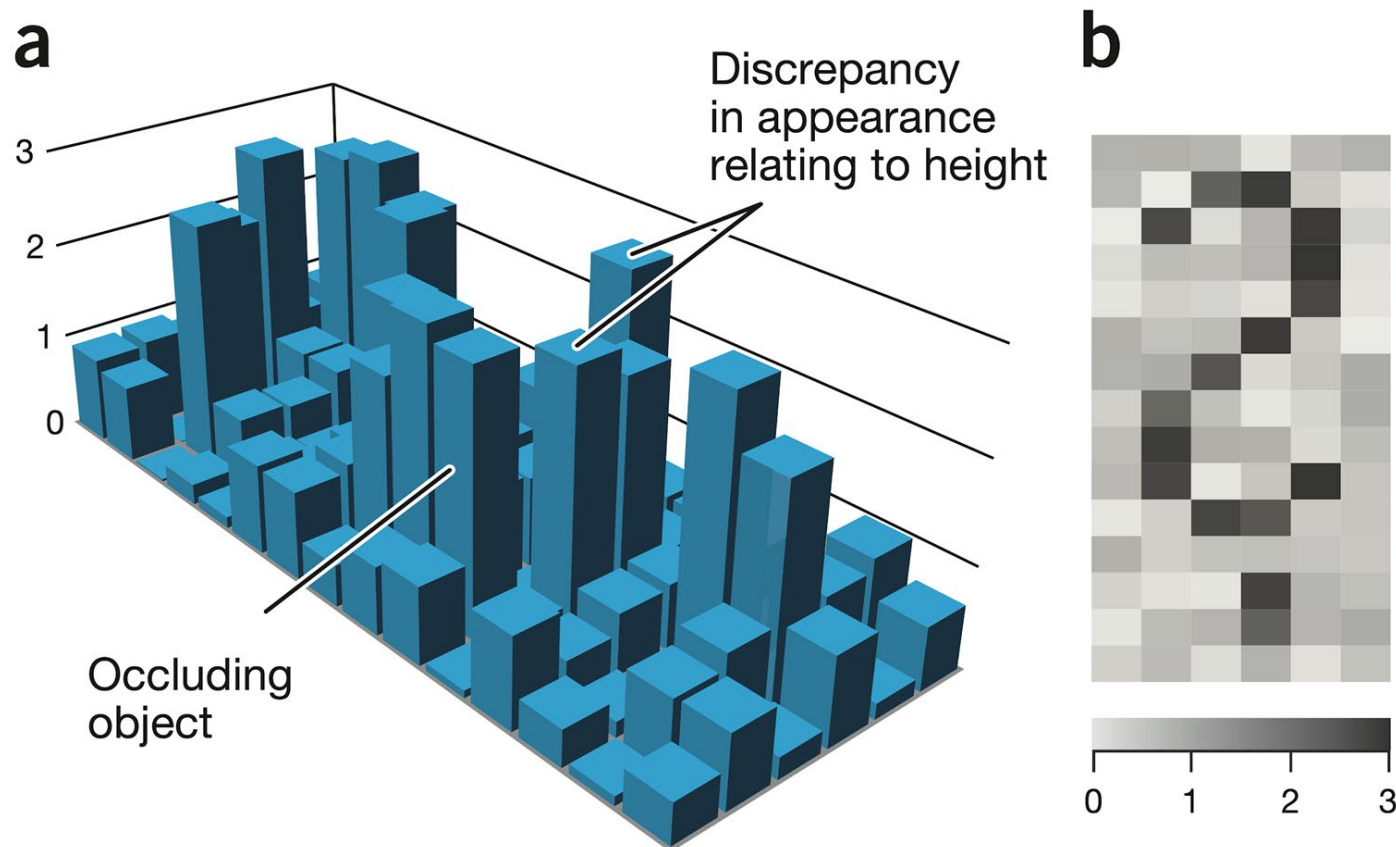
Visualizing Data in 3D ???

Stick to 2D whenever possible.



Visualizing Data in 3D ???

Stick to 2D whenever possible.



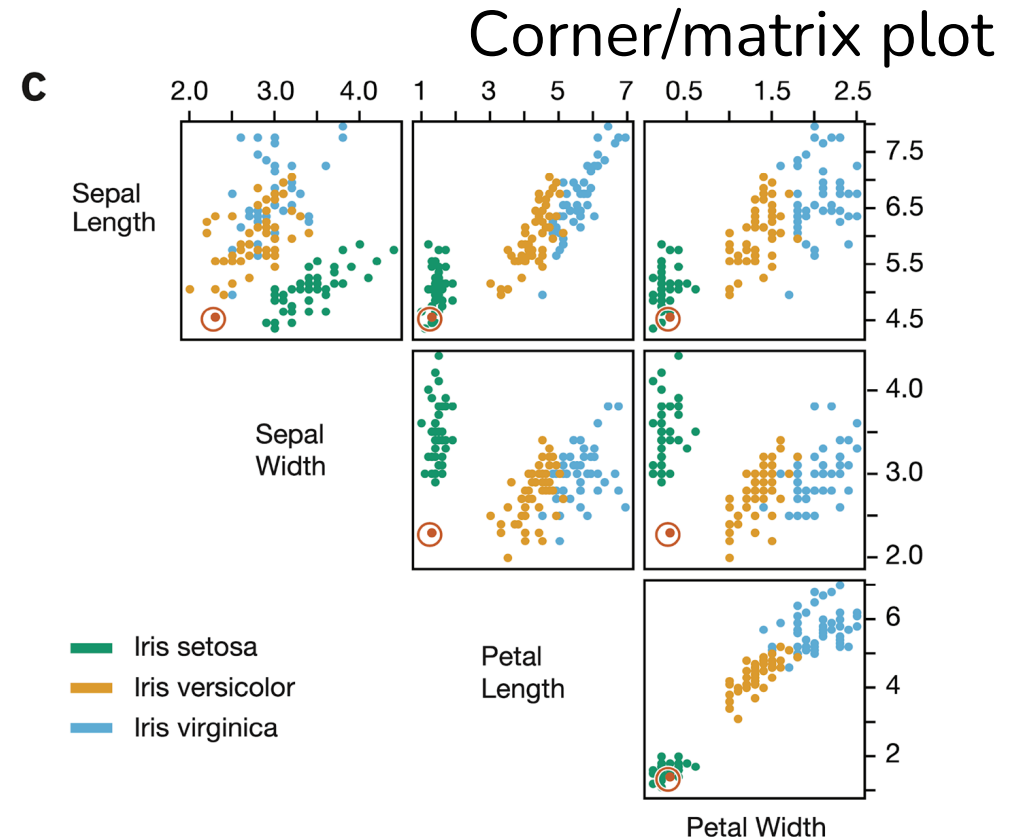
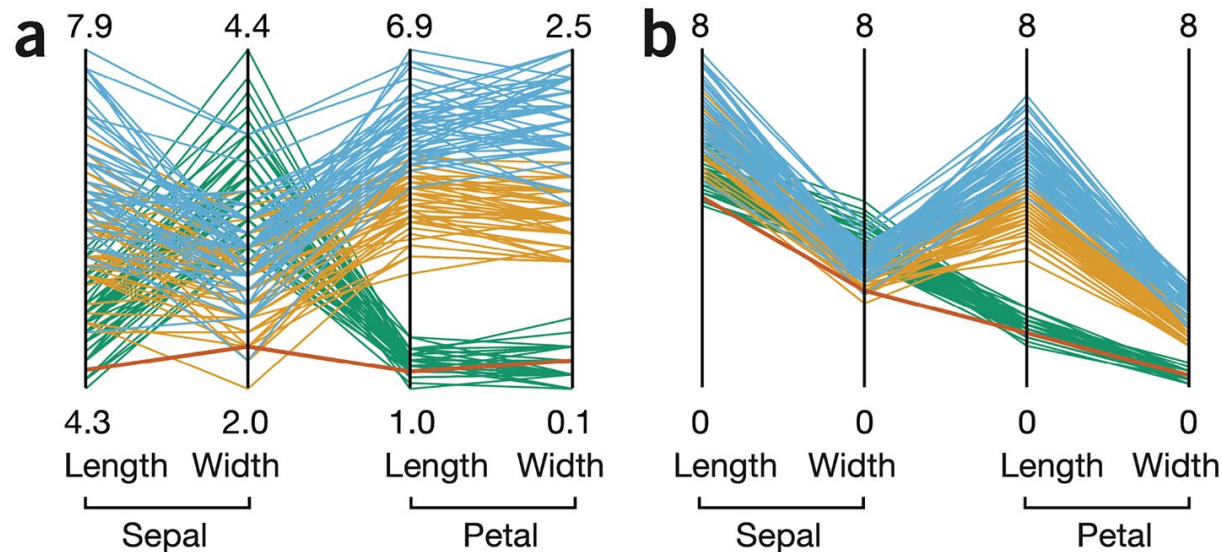
[Gehlenborg & Wong \(2012\)](#)

Visualizing Data in 3D ???

Stick to 2D whenever possible.

Combine multiple dimensions in 2D.

Parallel coordinate plots



Visualizing Data in 3D ???

When do you actually “need” 3D?

- Data has a genuine 3D shape that is missed in 2D (fields, asymmetric objects, etc.)
- Rotation reveals (hidden) structure

Strategies for successful 3D (on a 2D screen)

- **Interactivity is essential** (e.g., rotate, responsive)
- Add depth cues (floor grid, reference plane, drop lines, shadows, shading)
- Link 3D to 2D projections

Challenges

- Occlusion (near hides far)
- Perspective distortion (near has larger apparent size)
- **Manual data selection is hard!**

Firefly: Motivation

Originally to explore FIRE data

- Huge multi-dimensional data sets of galaxy simulations
- No available solution (at the time) to visualize, fly through, filter in real time
- Enable serendipitous discovery
- Firefly software had a few iterations; ended up as a browser-based tool

Later generalized to enable “any” 3D particle-based data

- Read multiple different data types
- Additional Python integration (e.g., data selection)
- Option to progressively load data (when dataset is larger than RAM)

Firefly

<https://firefly.rcs.northwestern.edu/>,

<https://ui.adsabs.harvard.edu/abs/2023ApJS..265...38G/abstract>

- Browser-based interactive particle viewer, capable of rendering millions of points in real-time (and to explore larger datasets with octree)
- Javascript frontend (three.js, d3.js, etc.)
- Python backend (Flask)
- Additional Python tools to format data for Firefly
- Built primarily by Alex Gurvich and Aaron Geller
 - mostly pre-2021 = before AI coding assistants ;)
- Example notebooks here :

<https://github.com/ageller/Firefly/tree/main/src/firefly/ntbks>



Firefly:

A WebGL tool to explore particle-based data

Aaron Geller / Alex Gurvich / Northwestern

Instructions:

- Right-click and drag with the mouse to rotate your view.
- Use the mouse wheel to zoom in and out.
- Click the Controls bar on the top left to show/hide a user interface.
- Detailed instructions can be found on [the Firefly GitHub page](#).
- **h**: toggles this help screen on and off.



Click to begin.

Gaia DR3 visualized using Firefly

Aaron Geller

Alex Gurvich



        Run   Code  

Try looking at this in Firefly

```
In [29]: # some_file.py
import sys
sys.path.insert(0, '/Users/ageller/Visualizations/Firefly/dataReader')

import pandas as pd
from simpleReader import simpleReader
import requests
from IPython.display import IFrame
```

Start the server

```
In [30]: %%bash --bg --proc bg_proc
python /Users/ageller/Visualizations/Firefly/FireflyFlaskApp.py
```

Show Firefly in an IFrame

```
In [ ]: url = "http://localhost:5000/combined"
IFrame(url, width=1650, height=700)
```

Format the data and send it to Firefly

```
In [32]: mx, my, mz = np.mean(resampleX), np.mean(resampleY), np.mean(resampleZ)
sx, sy, sz = np.std(resampleX - mx), np.std(resampleY - my), np.std(resampleZ - mz)
s = max([sx,sy,sz])
coords = np.array([(resampleX - mx)/s, (resampleY - my)/s, (resampleZ - mz)/s]).T
print(coords.shape)

name = "part1"
outDict = simpleReader(name, coords, [1,1,0,0.7], 1, colorArray=resampleRGBA)
print(outDict['parts'][name]['Coordinates'].shape)
```

Firefly

Time for you to try on your own machine.

I recommend creating a conda environment for firefly:

```
conda create --name firefly-env python=3.11 jupyter  
conda activate firefly-env  
pip install firefly
```

We'll start with `minimal_firefly_example.ipynb`

Questions?



Exercise

1. Download/generate/use your own data that has x,y,z and other attributes.
 - *Suggestion:* Gaia data (e.g., for a galactic open cluster)
2. Ingest this into Firefly and explore it in a Jupyter notebook.
 - Make sure to include attributes to color/filter by.
3. Use the data selection tool to select a portion of your data using Firefly.
4. Create a plot of these selected data using Python.
5. Share the result with us!